

Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

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1 **Research Highlights**

- 2 • One read operator over atom spaces covers attention, retrieval, and MoE
3 routing.
- 4 • A Lipschitz insertion condition and an exact tail certificate are shown to be
5 far too loose in practice.
- 6 • Anti-atom cancellation is exact but collapses under key perturbations of or-
7 der 0.05.
- 8 • Held-out ROC calibration lifts CounterFact-500 paraphrase activation from
9 38% to 53%.
- 10 • Matched GRACE run: 99–100% rewrite efficacy but 0–6% paraphrase acti-
11 vation on GPT-2.

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